Benchmarking and Learning Multi-Dimensional Quality Evaluator for Text-to-3D Generation

Yujie Zhang[1*] Bingyang Cui[1*] Qi Yang[2] Zhu Li[2] Yiling Xu[1†]
[1] Shanghai Jiao Tong University [2] University of Missouri-Kansas City
[1]{yujie19981026, ccccby0813, yl.xu}@sjtu.edu.cn , [2]{qiyang, lizhu}@umkc.edu

Abstract

Text-to-3D generation has achieved remarkable progress in recent years, yet evaluating these methods remains challenging for two reasons: i) Existing benchmarks lack fine-grained evaluation on different prompt categories and evaluation dimensions. ii) Previous evaluation metrics only focus on a single aspect (e.g., text-3D alignment) and fail to perform multi-dimensional quality assessment. To address these problems, we first propose a comprehensive benchmark named MATE-3D. The benchmark contains eight well-designed prompt categories that cover single and multiple object generation, resulting in 1,280 generated textured meshes. We have conducted a large-scale subjective experiment from four different evaluation dimensions and collected 107,520 annotations, followed by detailed analyses of the results. Based on MATE-3D, we propose a novel quality evaluator named HyperScore. Utilizing hypernetwork to generate specified mapping functions for each evaluation dimension, our metric can effectively perform multi-dimensional quality assessment. HyperScore presents superior performance over existing metrics on MATE-3D, making it a promising metric for assessing and improving text-to-3D generation. The project is available at https://mate-3d.github.io/.

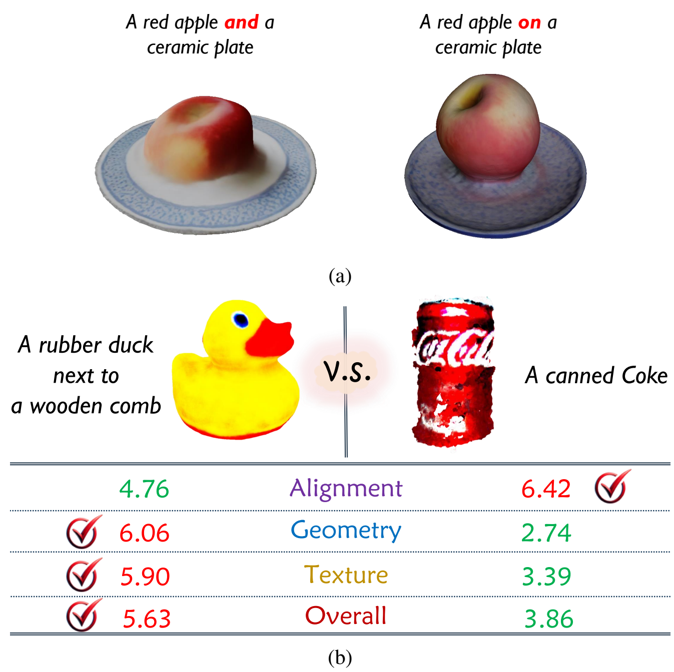


Figure 1. (a) Diverse generation results from similar prompts. (b) Quality evaluation from multiple perspectives.

1. Introduction

Remarkable advances in text-to-3D generative methods have been witnessed in recent years [33]. Given textual descriptions (*i.e.*, prompts), these methods have the capability to generate contextually relevant 3D representations, such as textured mesh [56], Neural Radiance Field (NeRF) [15, 21, 39], and 3D Gaussian Splatting (3DGS) [25, 48, 55, 71]. Unlike traditional distortions in 3D data processing (*e.g.*, noise, lossy compression), generated 3D representations may suffer from some unique degradations, such as misalignment with prompt semantics and multi-view inconsistency (*i.e.*, the Janus problem) [18, 44]. To better guide the method design and support fair comparisons, it is important to explore subjective-aligned quality evaluation for text-to-3D generation methods.

Previous works have conducted text-to-3D evaluation by proposing new benchmarks. For example, T^3Bench [14] constructs three prompt classes with increasing scene complexity, and *Multiple objects*, followed by collecting 630 texture meshes annotated from two dimensions, *i.e.*, quality and alignment. However, these benchmarks have two main drawbacks. i) **Lack of Prompt Diversity**. Although these benchmarks [14, 22, 61] have classified prompts based on textual complexity or object quantity, their categorization generally lacks an adequate degree of distinction. In fact,

* Equal Contribution.
† Corresponding Author.

we observe that some similar prompts usually lead to visually different results, as shown in Fig. 1a. Therefore, a fine-grained categorization is essential to gain a comprehensive understanding of model capabilities. ii) **Limited Evaluation Dimension.** As shown in Fig. 1b, the overall quality of the generated 3D representation is influenced by different factors, including geometry fidelity, texture details, and text-3D alignment. A single score is insufficient to perform detailed comparison between different cases. Consequently, it is necessary to construct a comprehensive benchmark with higher prompt diversity and richer evaluation dimensions.

Another challenge in text-to-3D evaluation is the lack of accurate and automated metrics to assess the quality of generated results. Existing approaches [59] rely primarily on subjective user studies, which are constrained by high costs in terms of time, money, and rigorous testing environment. Some previous work [40, 56, 64] has attempted to address this by rendering the 3D objects into single or multiple 2D images and using CLIP- [45] or BLIP- [29] based metrics [16, 49, 62, 65] to measure the alignment between the prompts and the rendered images. However, these metrics focus on the text-3D correspondence, ignoring other critical aspects of subjective perception, such as geometry and texture fidelity. In fact, human subjects can dynamically adjust their focus and decision process based on the evaluation dimension (*e.g.*, prioritize shape and contour for geometry evaluation), resulting in diverse scores with changing perspectives. Consequently, single-dimensional evaluators may fail to capture rich quality-related factors. To comprehensively evaluate text-to-3D generation, it is essential to develop robust metrics that reflect multi-dimensional human perception.

To solve the above problems, we establish a comprehensive benchmark named **M**ulti-Dimension**A**l **T**ext-to-3D Quality **E**valuation Benchmark (MATE-3D), which is the first contribution of this paper. MATE-3D first categorizes the prompts based on object quantity. For single object generation, we define four sub-categories based on the prompt complexity and creativity, including *"Basic"*, *"Refined"*, *"Complex"* and *"Fantastical"*. For multiple object generation, the prompts contain at least two objects with specified relationships between the objects. Based on the type of relationships, we define four sub-categories denoted by *"Grouped"*, *"Spatial"*, *"Action"*, and *"Imaginative"*. Utilizing large language models, we design 160 prompts for the eight sub-categories in total. We use these prompts as the input of eight prevalent text-to-3D methods and acquire 1,280 generated samples, which are transformed into a unified representation, *i.e.,* textured mesh. Then, we conduct a comprehensive subjective experiment and each textured mesh is annotated by 21 human subjects from four evaluation dimensions, including *semantic alignment*, *geometry quality*, *texture quality*, and *overall quality*. A total of $1,280 \times 4 \times 21 = 107,520$ annotations are collected, and we further obtain the Mean Opinion Score (MOS) for each evaluation dimension of the generated meshes. Finally, we conduct detailed analyses of subjective scores to provide useful insight for future text-to-3D research.

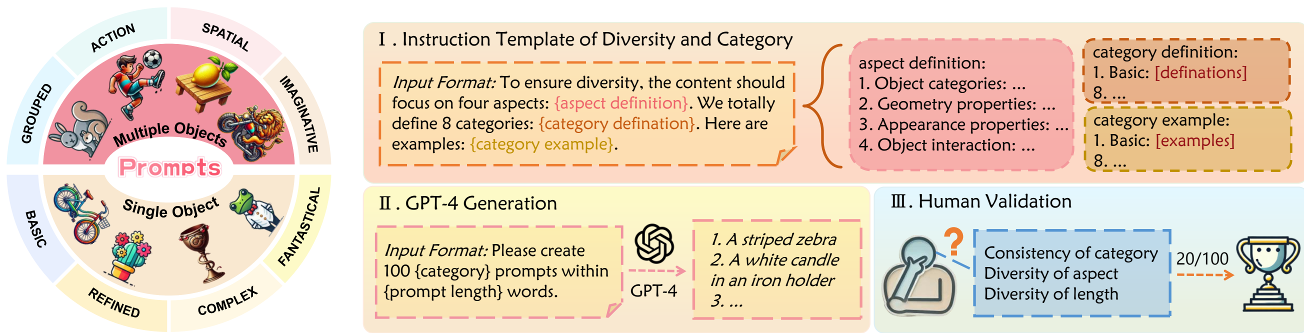
Furthermore, we propose a novel quality evaluator named HyperScore for text-to-3D generation, which can perform multi-dimensional assessment by utilizing a hypernetwork to generate specified mapping function for each evaluation dimension, resulting in the second contribution. More concretely, we first use a pre-trained CLIP model to extract visual and textual features from rendered images of textured meshes and prompts. Then, we create a sequence of learnable tokens and feed them into the textual encoder to obtain multiple condition features that represent information relevant to evaluation dimensions. Using these condition features, we implement two strategies to achieve fine-grained quality evaluation across dimensions. i) We utilize the condition features to compute the fusion weights for various patches in the visual feature, which mimics the focus shift during evaluation. ii) We feed the condition features into a hypernetwork [12] to generate diverse weights of a mapping head, thereby simulating different decision-making processes aligned with changing evaluation dimensions. Experiments on MATE-3D show that HyperScore outperforms existing metrics in all evaluation dimensions. We summarize the contributions as follows:

- We establish a new benchmark, MATE-3D, for evaluating text-to-3D methods. MATE-3D contains 1,280 textured meshes generated from eight prompt categories and each sample is annotated from four evaluation dimensions.
- We propose a multi-dimensional quality evaluator, HyperScore, for text-to-3D generation. Utilizing the learnable condition features, we achieve differentiated quality predictions for multiple evaluation dimensions.
- Extensive experiments show that the proposed HyperScore presents superior performance on different evaluation dimensions than existing evaluation metrics.

2. Related Works

2.1. Text-to-3D Generation

Generating 3D objects conditioned on texts has become a challenging yet prominent research area in recent years. With the emergence of large-scale 3D datasets [4, 5], some methods, such as Point-E [41] and Shap-E [24], first train the network using 3D datasets and then generate 3D assets in a feed-forward manner [1, 10, 11, 16, 19, 73, 74]. Since the quantity of image-text pairs [27, 28] is much larger than the 3D counterparts, Dreamfusion [44] pioneers the paradigm of optimizing 3D generation with pre-trained 2D diffusion models, which leverages Score Distillation Sam-



(a) (b)

Figure 2. (a) The overview of eight prompt categories; (b) The illustration of prompt generation pipeline.

pling (SDS) to optimize a NeRF. Based on this paradigm, some works [31, 37, 38, 51, 59, 77] introduce a coarse-to-fine optimization strategy with two stages and improve both speed and quality. Recent studies [23, 48, 55, 71, 72] incorporate 3DGS into generative 3D content creation, achieving acceleration compared to NeRF-based approaches. There also emerge hybrid 3D generative methods [2, 3, 7, 30, 32, 35, 50, 52, 54, 66–68, 70] by combining the advantage of 3D native and 2D prior-based generative methods. The typical example is One-2-3-45++ [34], which generates 3D representations by training a 3D diffusion model with the input of 2D prior-based multi-view images.

2.2. Text-to-3D Evaluation

To facilitate the text-to-3D evaluation, several benchmarks have been developed. T^{3}Bench [14] devises three prompt suites incorporating diverse 3D scenes and with increasing complexity, including *Single object*, *Single object with surroundings*, and *Multiple objects*, and then creates 630 samples scored from two dimensions, *i.e.,* quality and alignment. Another benchmark [61] leverages GPT-4 to generate prompts with varying degrees of creativity and complexity, then evaluates preferences between 234 sample pairs generated from identical prompts across five dimensions. Overall, these benchmarks face limitations, such as insufficient prompt diversity and the absence of multi-dimensional absolute scores, emphasizing the need for a more comprehensive benchmark. Furthermore, existing evaluation metrics [16, 26, 29, 49, 62, 65, 69] are either limited to single-dimensional evaluations or dependent on simple pairwise comparisons, which are insufficient to face the nuanced challenges of text-to-3D generation. Consequently, there is an urgent need for a comprehensive metric capable of assessing multi-dimensional quality.

There exists some work targeted for multi-dimensional evaluation in text-to-image generation. An intriguing way, MPS [75], introduces the condition features upon CLIP model to learn diverse dimensions. We inherit the merits of MPS and further introduce two effective strategies to achieve a fine-grained quality evaluation across dimensions.

3. Benchmark Construction and Analysis

3.1. Data Preparation

Prompt Categorization. To enhance the comprehensiveness and diversity of constructed prompts, we thoroughly consider the variety of scenarios present in the real world. We first classify text prompts based on object quantity. For single object generation, we define four sub-categories including *"Basic"*, *"Refined"*, *"Complex"* and *"Fantastical"*, where the former three categories have increasing textual complexity while the last category focuses on generating object with high creativity. For multiple object generation, the text prompt is composed of at least two objects with specified relationships between the objects. Based on the type of the relationships, we define four sub-categories denoted by *"Grouped"*, *"Action"*, *"Spatial"*, and *"Imaginative"*. The *Grouped* category describes multiple objects connected by "and"; The *Action* and *Spatial* categories describe action interactions (*e.g.*, "hold", "watch", "play with") and spatial (*e.g.* "on", "near", "on the bottom of"), respectively; *"Imaginative"* describe multiple objects with interactions, where objects or interactions should be creative. Fig. 3 shows eight exemplary prompts corresponding to different categories. We detail the definition of each category in the supplementary material.

Prompt Generation. Based on the defined categories, we generate well-structured prompts for each category with the assistance of GPT-4 [42] as shown in Fig. 2b. First, we follow [61] to provide an instruction to ensure that GPT-4 understands the task requirements. To promote prompt diversity, the instruction first emphasizes four aspects to consider: object categories, geometry properties, appearance properties, and object interactions. Then, the instruction defines the above eight categories with examples to ensure that the prompts generated by GPT-4 closely align with each category. Based on the instruction, we utilize GPT-4 to create a large pool of candidate prompts tailored to the identified challenges. Finally, we manually filter out any duplicate or inappropriate prompts to ensure consistency and diversity

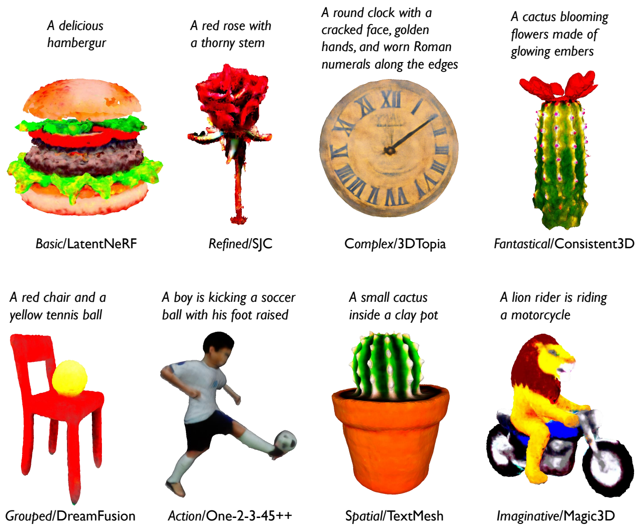


Figure 3. Samples from the database generated by eight different generative methods.

within the benchmark. Overall, we select 20 prompts from a pool of 100 candidates for each category, resulting in a total of 160 prompts.

Mesh Generation. After the prompt generation, we adopt eight recent text-to-3D generative methods, including DreamFusion [44], Magic3D [31], Score Jacobian Chaining (SJC) [57], TextMesh [56], 3DTopia [17], Consistent3D [63], LatentNeRF [38], and One-2-3-45++ [34] to generate 3D representations by using open source code and default weight, which are then transformed into textured meshes. Finally, we obtain 1,280 textured meshes. Fig. 3 shows the mesh samples in the database along with their corresponding prompts and generative methods.

3.2. Subjective Study

Evaluation Dimension. The quality of the generated textured meshes is influenced by different factors. To provide a more nuanced understanding of human perception, we define four evaluation dimensions that need annotation:

- **Alignment.** Subjects should measure the semantic consistency of the generated meshes with the prompts, including evaluating whether the generated meshes accurately match the prompts (*e.g.*, quantity, attributes, location, relationship) and whether there is missing or redundant content.
- **Geometry.** Subjects should measure the geometry fidelity of textured meshes, including measuring the shape, contour, size, and whether there is missing or redundant structure in the generated meshes.
- **Texture.** Subjects should measure the texture details of textured meshes, including evaluating the color, material, and whether the appearance of meshes is vibrant and of high resolution.
- **Overall.** Subjects should conduct a holistic assessment of the samples by integrating the insights from different perspectives.

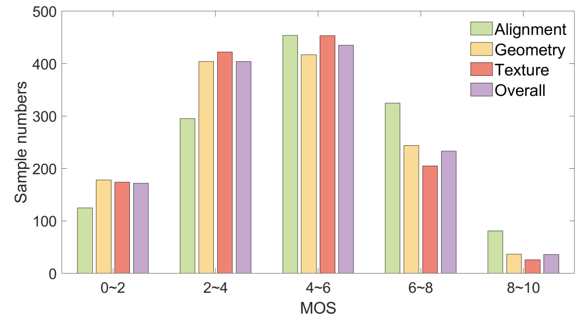


Figure 4. Distribution of MOS for four dimensions.

Experiment Setup. To obtain the MOS of each mesh, we invite 21 subjects to score the samples from the above four dimensions. We employ the 11-level impairment scale (ranging from 0 to 10) proposed by ITU-T P.910 [47] as the voting methodology. After collecting the subjective scores, we implement the outlier detection method outlined in ITU-R BT.500 [46] to identify and exclude outliers from the raw data. Finally, the MOS for each dimension of one textured mesh is calculated by averaging the filtered raw scores. Fig. 4 shows the distribution of MOS for alignment, geometry, texture, and overall quality. We see that the benchmark covers broad quality ranges for multiple dimensions.

3.3. Observations

We conduct comprehensive analyses for the subjective scores of MATE-3D. We first investigate the relationships among different evaluation dimensions. Fig. 5a - Fig. 5c illustrate scatter plots that depict the correlations among three dimensions (*i.e.*, alignment, geometry, and texture). From Fig. 5a and Fig. 5b, we observe a relatively low correlation between the alignment and the geometry or texture score at the medium quality range (*e.g.*, $[4, 6]$). The reason is that the alignment evaluation mostly focuses on semantic consistency while the geometry or texture evaluation relies on specified details. For instance, in Fig. 5a, the mesh generated from the prompt "A plastic spoon and a ceramic cup" includes only a cup, resulting in a low alignment score. However, the cup itself demonstrates good geometry quality. In contrast, the sample corresponding to the prompt "A canned Coke" achieves a relatively high alignment score but suffers from an uneven surface and an incomplete shape, leading to a low geometry score. From Fig. 5c, the mutual influence between the geometry and texture scores is more pronounced, as the two dimensions influence each other during the generation process. Finally, in Fig. 5d, we calculate the deviations between the other three dimensions and the overall score after non-linear re-

Table 1. Average scores of each generative method on eight prompt categories. The values of 1-8 represent the rank in each row.

Method	Single Object				Multiple Object			
	Basic	Refined	Complex	Fantastic	Grouped	Action	Spatial	Imaginative
DreamFusion	4.99(1)	4.96(2)	4.64(4)	3.24(8)	3.45(7)	3.73(6)	4.47(3)	3.75(5)
Magic3D	5.26(1)	4.89(6)	5.06(3)	4.95(4)	3.92(8)	4.73(7)	5.06(2)	4.93(5)
SJC	3.28(2)	3.48(1)	2.95(4)	2.92(6)	2.50(8)	2.77(7)	2.92(5)	3.25(3)
TextMesh	4.08(6)	4.56(2)	4.79(1)	4.22(4)	3.22(8)	3.68(7)	4.17(5)	4.26(3)
3DTopia	5.06(1)	4.91(2)	4.83(3)	3.89(6)	4.05(4)	2.80(8)	3.89(5)	2.95(7)
Consistent3D	4.15(5)	4.92(1)	4.40(2)	4.32(4)	3.37(8)	3.64(6)	4.39(3)	3.47(7)
LatentNeRF	3.20(7)	3.76(3)	3.47(5)	4.16(1)	2.94(8)	3.48(4)	3.46(6)	4.01(2)
One-2-3-45++	7.79(1)	7.69(2)	6.50(5)	6.60(4)	6.49(6)	5.65(8)	6.81(3)	6.13(7)

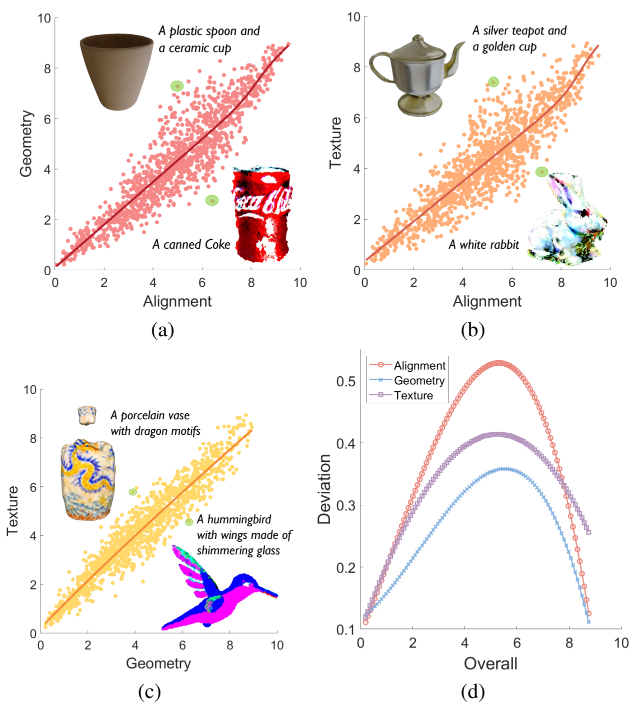


Figure 5. (a) Alignment *vs.* Geometry; (b) Alignment *vs.* Texture; (c) Geometry *vs.* Texture; (d) Deviations among the other three dimensions and overall quality.

gression, which indicates the consistency between each dimension and the overall quality. We can see that the geometry quality is most closely related to the overall quality. It is reasonable because geometry deformations (*e.g.*, incomplete shape, floaters) directly affect 3D perception. The alignment quality appears to have the least correlation with the overall quality because it relatively ignores some details that human subjects care about.

We further illustrate the average scores of each generative method on the four evaluation dimensions in Fig. 6. We can see that One-2-3-45++ performs best on all dimensions, while SJC provides the worst performance. This is primarily because SJC usually causes an incomplete geometry shape and noisy floaters, making it difficult to generate high-quality samples. Other generative methods also face some difficulties during the generation process. For example, TextMesh often generates 3D objects with overly simplistic shapes, which limits their complexity and realism. 3DTopia tends to generate single object, leading to comparatively lower scores in multiple object generation. One-2-3-45++ tends to generate flatter geometric structures and blurry textures. Moreover, except for One-2-3-45++, all generative methods can cause Janus problem due to the bias of the used 2D diffusion models towards 2D front views. The bias results in repeating front views from different angles rather than generating coherent 3D objects.

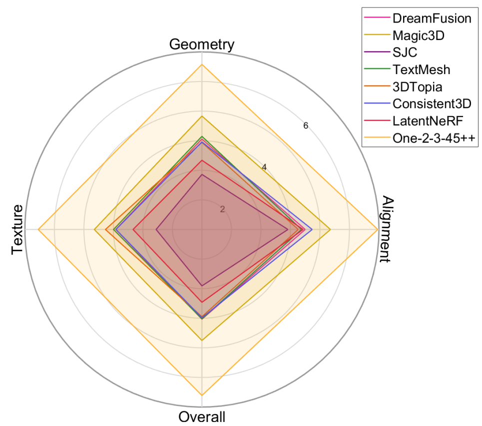


Figure 6. Average scores of different methods on four dimensions.

To investigate the ability of the methods to generate different scenes, we report the average scores (in terms of overall quality) of each method on eight prompt categories in Tab. 1. We have the following findings: i) All methods achieve better performance for single object generation than multiple object generation. In fact, existing methods struggle in constructing the correct relationships among objects and usually miss some contents during generation, restricting their performance in multiple object generation. ii) For the sub-categories of single object generation, the generation quality of most methods decreases with increasing prompt complexity (from *Basic* to *Refined* to *Complex*). However, there are some exceptions. For example, the gen-

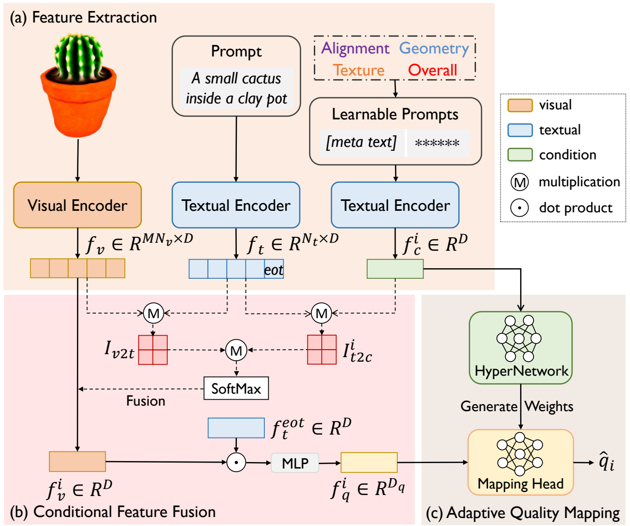


Figure 7. The framework of the proposed evaluator.

erated samples of TextMesh often exhibit too simple geometric structures. Consequently, its generation for the *Basic* category may be too simplistic to discern, leading to lower scores. iii) For the sub-categories of multiple object generation, it is observed that most methods provide better performance on the *Spatial* category, followed by the *Action* and *Grouped* categories. It seems that these methods are more capable of modeling specific spatial interactions. In contrast, the *Action* prompts may introduce ambiguity with actions like "chase" or "run" because these verbs often indicate a greater distance, while the *Grouped* category usually results in overlapping or missing objects. iv) The *Fantastic* and *Imaginative* categories primarily emphasize the creativity of the prompts, allowing a wide range of object details and interactions. As a result, different methods perform inconsistently on the two categories.

4. Evaluator for Text-to-3D Generation

4.1. Problem Formulation

Given a generated mesh x and its corresponding prompt t, existing single-dimensional evaluators directly map them to the subjective score q. The procedure can be described as $\hat{q} = \psi(\phi(x,t))$, where $\hat{q}$ is the predicted score; $\phi(\cdot)$ denotes a feature extractor such as CLIP and $\psi(\cdot)$ represents a mapping function such as the cosine similarity function in CLIPScore [16]. Given multiple targets $\{q_i\}_{i=1}^{K}$ from K dimensions, a direct approach is to separately use a single-head network for each dimension. However, it is not economical to train and retain dozens of different expert models. Therefore, we prefer a one-for-all evaluator. For the purpose, a solution is a multi-task network with a shared feature extractor and multiple mapping heads, that is,

$$\hat{q}_i = \psi_i\left(\phi\left(x,t\right)|\theta_i\right), \tag{1}$$

where $\psi_i(\cdot|\theta_i)$ denotes the mapping head for the i-th dimension with the parameters θ_i. However, directly optimizing such a multi-task network may fail to exploit differentiated perception rules depending on the evaluation dimensions. Therefore, we further transform the task as follows:

$$\hat{q}_i = \psi\left(\phi\left(x,t\right)|\pi\left(f_c^i\right)\right), \tag{2}$$

where $\pi(\cdot)$ denotes a hypernetwork to generate θ_i for a unified mapping head; f_c^i denotes a condition feature related to the i-th evaluation dimension. By injecting the condition features, we can better characterize the differences among evaluation perspectives.

We illustrate the framework of our evaluator in Fig. 7, which can be divided into three stages: i) **Feature Extraction**. We first use a transformer-based CLIP model to extract visual and textual features from rendered images of textured meshes and prompts. Meanwhile, we feed a sequence of learnable prompts into the textual encoder to obtain multiple condition features corresponding to various dimensions. ii) **Conditional Feature Fusion**. We utilize the condition features to compute fusion weights for various patches in the visual features. The fused visual feature is then merged with the textual feature to obtain the final quality feature. iii) **Adaptive Quality Mapping**. We feed the condition features into a hypernetwork to generate diverse weights of the mapping head, and then regress the quality feature into the final score of the specified dimension.

4.2. Feature Extraction

For a test mesh x, we first render it into multi-view images as $\{x_r^m\}_{m=1}^{M}$, where M denotes the number of viewpoints. Then we resort to a pre-trained CLIP model for feature extraction, which is composed of a transformer-based visual encoder $\phi_v(\cdot)$ and a frozen textual encoder $\phi_t(\cdot)$. To better capture local distortion patterns, we use the tokens of all image patches as the output of $\phi_v(\cdot)$. The feature extraction process can be formulated as:

$$f_v^m = \phi_v\left(x_r^m\right) \in \mathbb{R}^{N_v \times D},\ f_t = \phi_t\left(t\right) \in \mathbb{R}^{N_t \times D}, \tag{3}$$

where N_v denotes the token number of image (*i.e.,* the patch number) and N_t denotes the token number of text. We further concatenate the visual features of different viewpoints to derive $f_v \in \mathbb{R}^{MN_v \times D}$.

Previous research [20, 75] demonstrates that specific tags (*e.g.,* "shape, edge" for geometry evaluation) can effectively capture dimension-related information. To avoid meticulous design for fixed tags, we only define K meta text about different evaluation dimensions (*e.g., "alignment quality", "geometry quality", "texture quality", "overall quality"* for MATE-3D) and acquire the corresponding meta tokens. We further introduce the learnable prompts following CoOp [76] and insert the meta tokens in the front of a squeeze of

Table 2. Performance Comparison on MATE-3D. The best metric in each column is marked in **boldface**.

Metric	Alignment			Geometry			Texture			Overall		
	PLCC	SRCC	KRCC	PLCC	SRCC	KRCC	PLCC	SRCC	KRCC	PLCC	SRCC	KRCC
CLIPScore [16]	0.535	0.494	0.347	0.520	0.496	0.347	0.563	0.537	0.379	0.541	0.510	0.359
BLIPScore [29]	0.569	0.533	0.377	0.570	0.542	0.382	0.608	0.578	0.413	0.586	0.554	0.393
Aesthetic Score [49]	0.185	0.099	0.066	0.270	0.160	0.107	0.288	0.172	0.114	0.228	0.150	0.100
ImageReward [65]	0.675	0.651	0.470	0.624	0.591	0.422	0.650	0.612	0.441	0.657	0.623	0.448
DreamReward [69]	0.533	0.526	0.368	0.528	0.501	0.349	0.544	0.513	0.359	0.551	0.524	0.367
HPS v2 [62]	0.477	0.416	0.290	0.455	0.412	0.286	0.469	0.423	0.296	0.465	0.420	0.293
CLIP-IQA [58]	0.151	0.039	0.025	0.223	0.125	0.084	0.264	0.163	0.108	0.213	0.115	0.076
Q-Align [60]	0.223	0.199	0.135	0.389	0.355	0.300	0.371	0.431	0.300	0.371	0.340	0.234
ResNet50 + FT [13]	0.577	0.551	0.387	0.669	0.653	0.471	0.691	0.672	0.487	0.649	0.632	0.452
ViT-B + FT [8]	0.544	0.515	0.360	0.662	0.642	0.461	0.691	0.676	0.492	0.633	0.614	0.439
SwinT-B + FT [36]	0.534	0.506	0.359	0.626	0.610	0.438	0.654	0.640	0.465	0.608	0.592	0.425
DINO v2 + FT [43]	0.664	0.642	0.461	0.761	0.739	0.550	0.787	0.771	0.579	0.746	0.728	0.538
MultiScore	0.657	0.638	0.458	0.719	0.703	0.516	0.746	0.729	0.540	0.714	0.698	0.511
HyperScore	**0.754**	**0.739**	**0.547**	**0.793**	**0.782**	**0.588**	**0.822**	**0.811**	**0.619**	**0.804**	**0.792**	**0.600**

learnable tokens to acquire K tokenized prompts as $\left\{t_c^i\right\}_{i=1}^K$. Subsequently, these tokenized prompts will pass through the frozen textual encoder to obtain a list of condition features denoted by $\left\{f_c^i \in \mathbb{R}^D\right\}_{i=1}^K$, where we only retain the EOT (end of text) token as the output because it is often used to represent the whole text.

4.3. Conditional Feature Fusion

As aforementioned, human evaluators adaptively focus on different aspects of scenes depending on the evaluation dimensions. Therefore, different patches in multiple viewpoints do not contribute equally to the final prediction; meanwhile, the contribution of each patch may change across dimensions. Based on the insight, we implement a conditional feature fusion strategy.

Specifically, we first normalize the visual, textual and condition features to obtain $\tilde{f}_v$, $\tilde{f}_t$, and $\left\{\tilde{f}_c^i\right\}_{i=1}^K$. For each evaluation dimension, we define two correlation matrices as:

$$I_{v2t} = \tilde{f}_v \cdot \tilde{f}_t^T \in \mathbb{R}^{MN_v \times N_t},\ I_{t2c}^i = \tilde{f}_t \cdot \left(\tilde{f}_c^i\right)^T \in \mathbb{R}^{N_t}, \tag{4}$$

where I_{v2t} quantifies the degree of correlation between each image patch and each text token, whereas I_{t2c}^i determines the relative importance of each text token for the i-th condition. Then, we perform the conditional feature fusion for the i-th evaluation dimension as:

$$f_{v,c}^i = \mathrm{SoftMax}\left(I_{v2t} \cdot I_{t2c}^i\right) \cdot f_v \in \mathbb{R}^D, \tag{5}$$

where $\mathrm{SoftMax}(\cdot)$ denotes the softmax function. Based on Eq. (5), the more relevant the patch is to the condition, the higher the weight. Finally, we derive the quality feature of the i-th dimension as:

$$f_q^i = \mathrm{MLP}\left(f_{v,c}^i \odot f_t^{eot}\right) \in \mathbb{R}^{D_q}, \tag{6}$$

where $\odot$ denotes the element-wise multiplication and $\mathrm{MLP}(\cdot)$ represents a simple multi-layer perception; $f_t^{eot} \in \mathbb{R}^D$ indicates the EOT token in f_t.

4.4. Adaptive Quality Mapping

After obtaining the quality feature for the specified condition, we need to transform it into the corresponding quality score using a mapping function $\psi(\cdot)$ parameterized by the weights θ_i. Previous studies [53] have indicated that the mapping function can represent the decision procedure used to assess quality. We further argue that as the evaluation dimension considered varies, the decision procedure also varies. Therefore, we define $\psi(\cdot)$ as fully connected layers and use a hypernetwork $\pi(\cdot)$ to generate θ_i, *i.e.*, $\theta_i = \pi\left(f_c^i\right)$. Two types of network parameters, *i.e.*, fully connected layer weights and biases, are generated (please refer to the supplementary material for the detailed architecture of $\psi(\cdot)$ and $\pi(\cdot)$). Finally, we can infer the quality score of the i-th dimension as:

$$\hat{q}_i = \psi\left(f_q^i | \pi\left(f_c^i\right)\right). \tag{7}$$

5. Experiment

5.1. Implement Details

Training Strategy. We implement our metric by PyTorch and conduct training and testing on the NVIDIA 3090 GPUs. All textured meshes are rendered into $M = 6$ projected images with a spatial resolution of 512 × 512 by PyTorch3D. More details for network implementation can be found in the supplementary material.

Training-test Split. We apply a 5-fold cross-validation for MATE-3D while ensuring that there is no prompt overlap between the training and testing sets. Specifically, the ratios of prompts in the training set and test set are 128:32. For each fold, the performance on the test set with mini-

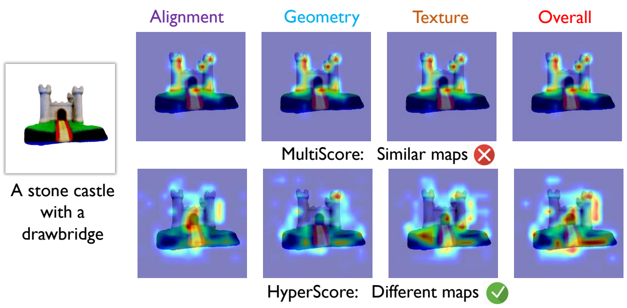


Figure 8. Visual explanations generated by XGrad-CAM for different evaluation dimensions.

mal training loss is recorded, and the average performance across all folds is recorded as the final result.

Evaluation Metrics. We utilize Pearson Linear Correlation Coefficient (PLCC), Spearman Rank order Correlation Coefficient (SRCC), and Kendall's Rank order Correlation Coefficient (KRCC) as performance criteria. Better metrics should have higher SRCC, KRCC, and PLCC.

5.2. Performance Comparison

Quantitative Analysis. Table 2 lists the experimental results on MATE-3D. We select widely used zero-shot metrics for text-to-image(or 3D) evaluation, including CLIPscore [16], BLIPScore [29], Aesthetic Score [49], ImageReward [65], HPS v2 [62], CLIP-IQA [58] and Q-Align [60]. Note that we average the scores of multiple rendered viewpoints for these metrics. In addition, we also equip some prevalent backbones including ResNet50 [13], ViT-B [8], SwinT-B [36] pre-trained on ImageNet [6] and ViT-B in DINO v2 [43] with multiple regression heads and fine-tune them on MATE-3D. Finally, we propose a baseline named MultiScore that has the same feature extractor as HyperScore but only applies a multi-task learning strategy.

From Tab. 2, we have the following observations: i) The proposed HyperScore exhibits outstanding performance across all evaluation dimensions, which demonstrates its effectiveness for multi-dimensional assessment. Especially, HyperScore presents a noticeable gain over MultiScore, justifying the potential of conditional learning strategy. ii) ImageReward performs best among these zero-shot metrics, which may be attributed to its rank-based training strategy. iii) Aesthetic Score provides inferior results, probably due to its preference for aesthetic quality. CLIP-IQA also performs poorly, possibly because they are designed for natural images instead of rendered images of textured meshes.

Visualization For better illustration, we present a qualitative analysis using a prevalent tool (*i.e.*, XGrad-CAM [9]) for network explanation. As illustrated in Fig. 8, the XGrad-CAM maps of MultiScore (top row) and HyperScore (bottom row) reflect what makes the network perform decisions for different evaluation dimensions. We can see that the multi-head network fails to distinguish different dimensions while HyperScore exhibits different focus areas corresponding to changes in evaluation dimensions, which validates the effectiveness of the proposed strategy. Notably, for HyperScore, the XGrad-CAM maps related to the geometry evaluation pay more attention to the shape and structure of objects. In contrast, the texture and overall quality evaluations have a wider range of focus, likely because they require more appearance details to make decisions.

Table 3. Ablation study for the key modules. '✓' or '✗' means the setting is preserved or discarded.

Index	CFF	AQM	Alignment	Geometry	Texture	Overall
(1)	✗	✗	0.638	0.703	0.729	0.698
(2)	✓	✗	0.660	0.730	0.760	0.724
(3)	✗	✓	0.721	0.762	0.792	0.776
(4)	✓	✓	**0.739**	**0.782**	**0.811**	**0.792**
Separately Trained			0.737	0.770	0.798	0.778

5.3. Ablation Study

We further investigate the contribution of different components and report the results (in terms of SRCC) in Tab. 3. In the table, CFF and AQM denote conditional feature fusion and adaptive quality mapping, respectively. The baseline (denoted by (1)) applies a multi-task learning strategy and simply averages all patches in the visual feature. *Separately Trained* denotes separately trained networks for each dimension. From the table, we can see: i) Comparing (2) or (3) with (1), both CFF and AQM contribute to the performance, demonstrating the effectiveness of the two modules. Especially, the AQM brings noticeable gain over the baseline, justifying the use of hypernetwork in our metric. ii) Seeing (2)-(4), the full model performs better than using only single module, which verifies the complementarity of the two modules. iii) Comparing (1), (4) and *Separately Trained*, directly training multi-head network results into performance drop compared to separately trained networks for each dimension while HyperScore can even achieve better performance, justifying the proposed strategy.

6. Conclusion

In this paper, we are dedicated to exploring a subjective-aligned quality evaluation for text-to-3D generation. For this purpose, we first establish a comprehensive benchmark named MATE-3D containing 1,280 textured mesh samples annotated from four evaluation dimensions. Based on MATE-3D, we propose a multi-dimensional quality evaluator named HyperScore. Based on the merits of hypernetwork, the evaluator can perform adaptive multi-dimensional prediction. Experimental results indicate that HyperScore is effective in evaluating text-to-3D generation.

Acknowledgments

This paper is supported in part by National Natural Science Foundation of China (62371290), National Key R&D Program of China (2024YFB2907204), the Fundamental Research Funds for the Central Universities of China, and STCSM under Grant (22DZ2229005).

References

[1] Anpei Chen, Haofei Xu, Stefano Esposito, Siyu Tang, and Andreas Geiger. Lara: Efficient large-baseline radiance fields. In *European Conference on Computer Vision (ECCV)*, 2025. 2

[2] Cheng Chen, Xiaofeng Yang, Fan Yang, Chengzeng Feng, Zhoujie Fu, Chuan-Sheng Foo, Guosheng Lin, and Fayao Liu. Sculpt3d: Multi-view consistent text-to-3d generation with sparse 3d prior. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[3] Yang Chen, Yingwei Pan, Haibo Yang, Ting Yao, and Tao Mei. Vp3d: Unleashing 2d visual prompt for text-to-3d generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[4] Matt Deitke, Ruoshi Liu, Matthew Wallingford, Huong Ngo, Oscar Michel, Aditya Kusupati, Alan Fan, Christian Laforte, Vikram Voleti, Samir Yitzhak Gadre, Eli VanderBilt, Aniruddha Kembhavi, Carl Vondrick, Georgia Gkioxari, Kiana Ehsani, Ludwig Schmidt, and Ali Farhadi. Objaverse-xl: A universe of 10m+ 3d objects. In *Advances in Neural Information Processing Systems (NIPS)*, 2023. 2

[5] Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsanit, Aniruddha Kembhavi, and Ali Farhadi. Objaverse: A universe of annotated 3d objects. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[6] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2009. 8

[7] Shaocong Dong, Lihe Ding, Zhanpeng Huang, Zibin Wang, Tianfan Xue, and Dan Xu. Interactive3d: Create what you want by interactive 3d generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[8] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 7, 8

[9] Ruigang Fu, Qingyong Hu, Xiaohu Dong, Yulan Guo, Yinghui Gao, and Biao Li. Axiom-based grad-cam: Towards accurate visualization and explanation of cnns. *arXiv preprint arXiv:2008.02312*, 2020. 8

[10] Rao Fu, Xiao Zhan, Yiwen Chen, Daniel Ritchie, and Srinath Sridhar. Shapecrafter: A recursive text-conditioned 3d shape generation model. In *Advances in Neural Information Processing Systems (NIPS)*, 2022. 2

[11] Jun Gao, Tianchang Shen, Zian Wang, Wenzheng Chen, Kangxue Yin, Daiqing Li, Or Litany, Zan Gojcic, and Sanja Fidler. Get3d: A generative model of high quality 3d textured shapes learned from images. In *Advances in Neural Information Processing Systems (NIPS)*, 2022. 2

[12] David Ha, Andrew M Dai, and Quoc V Le. Hypernetworks. In *International Conference on Learning Representations (ICLR)*, 2022. 2

[13] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 7, 8

[14] Yuze He, Yushi Bai, Matthieu Lin, Wang Zhao, Yubin Hu, Jenny Sheng, Ran Yi, Juanzi Li, and Yong-Jin Liu. T^3bench: Benchmarking current progress in text-to-3d generation. *arXiv preprint arXiv:2310.02977*, 2023. 1, 3

[15] Yuze He, Peng Wang, Yubin Hu, Wang Zhao, Ran Yi, Yong-Jin Liu, and Wenping Wang. Mmpi: a flexible radiance field representation by multiple multi-plane images blending. In *International Conference on Robotics and Automation (ICRA)*, 2024. 1

[16] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. CLIPScore: A reference-free evaluation metric for image captioning. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2021. 2, 3, 6, 7, 8

[17] Fangzhou Hong, Jiaxiang Tang, Ziang Cao, Min Shi, Tong Wu, Zhaoxi Chen, Tengfei Wang, Liang Pan, Dahua Lin, and Ziwei Liu. 3dtopia: Large text-to-3d generation model with hybrid diffusion priors. *arXiv preprint arXiv:2403.02234*, 2024. 4

[18] Susung Hong, Donghoon Ahn, and Seungryong Kim. Debiasing scores and prompts of 2d diffusion for view-consistent text-to-3d generation. In *Advances in Neural Information Processing Systems (NIPS)*, 2023. 1

[19] Yicong Hong, Kai Zhang, Jiuxiang Gu, Sai Bi, Yang Zhou, Difan Liu, Feng Liu, Kalyan Sunkavalli, Trung Bui, and Hao Tan. Lrm: Large reconstruction model for single image to 3d. In *International Conference on Learning Representations (ICLR)*, 2024. 2

[20] Jingwen Hou, Weisi Lin, Yuming Fang, Haoning Wu, Chaofeng Chen, Liang Liao, and Weide Liu. Towards transparent deep image aesthetics assessment with tag-based content descriptors. *IEEE Transactions on Image Processing (TIP)*, 2023. 6

[21] Yi-Hua Huang, Yan-Pei Cao, Yu-Kun Lai, Ying Shan, and Lin Gao. Nerf-texture: Texture synthesis with neural radiance fields. In *ACM SIGGRAPH*, 2023. 1

[22] Ka-Hei Hui, Aditya Sanghi, Arianna Rampini, Kamal Rahimi Malekshan, Zhengzhe Liu, Hooman Shayani, and Chi-Wing Fu. Make-a-shape: a ten-million-scale 3D shape model. In *International Conference on Machine Learning (ICML)*, 2024. 1

[23] Vishnu Jaganathan, Hannah Hanyun Huang, Muhammad Zubair Irshad, Varun Jampani, Amit Raj, and Zsolt Kira. Ice-g: Image conditional editing of 3d gaussian splats. *arXiv preprint arXiv:2406.08488*, 2024. 3

[24] Heewoo Jun and Alex Nichol. Shap-e: Generating conditional 3d implicit functions. *arXiv preprint arXiv:2305.02463*, 2023. 2

[25] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions On Graphics (TOG)*, 2023. 1

[26] Baiqi Li, Zhiqiu Lin, Deepak Pathak, Jiayao Li, Yixin Fei, Kewen Wu, Xide Xia, Pengchuan Zhang, Graham Neubig, and Deva Ramanan. Evaluating and improving compositional text-to-visual generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[27] Chunyi Li, Tengchuan Kou, Yixuan Gao, Yuqin Cao, Wei Sun, Zicheng Zhang, Yingjie Zhou, Zhichao Zhang, Weixia Zhang, Haoning Wu, Xiaohong Liu, Xiongkuo Min, and Guangtao Zhai. Aigiqa-20k: A large database for ai-generated image quality assessment. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024. 2

[28] Chunyi Li, Zicheng Zhang, Haoning Wu, Wei Sun, Xiongkuo Min, Xiaohong Liu, Guangtao Zhai, and Weisi Lin. Agiqa-3k: An open database for ai-generated image quality assessment. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2024. 2

[29] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International Conference on Machine Learning (ICML)*, 2022. 2, 3, 7, 8

[30] Ming Li, Pan Zhou, Jia-Wei Liu, Jussi Keppo, Min Lin, Shuicheng Yan, and Xiangyu Xu. Instant3d: Instant text-to-3d generation. *International Journal of Computer Vision (IJCV)*, 2024. 3

[31] Chen-Hsuan Lin, Jun Gao, Luming Tang, Towaki Takikawa, Xiaohui Zeng, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, and Tsung-Yi Lin. Magic3d: High-resolution text-to-3d content creation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 3, 4

[32] Fangfu Liu, Diankun Wu, Yi Wei, Yongming Rao, and Yueqi Duan. Sherpa3d: Boosting high-fidelity text-to-3d generation via coarse 3d prior. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[33] Jian Liu, Xiaoshui Huang, Tianyu Huang, Lu Chen, Yuenan Hou, Shixiang Tang, Ziwei Liu, Wanli Ouyang, Wangmeng Zuo, Junjun Jiang, et al. A comprehensive survey on 3d content generation. *arXiv preprint arXiv:2402.01166*, 2024. 1

[34] Minghua Liu, Ruoxi Shi, Linghao Chen, Zhuoyang Zhang, Chao Xu, Xinyue Wei, Hansheng Chen, Chong Zeng, Jiayuan Gu, and Hao Su. One-2-3-45++: Fast single image to 3d objects with consistent multi-view generation and 3d diffusion. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3, 4

[35] Yibo Liu, Zheyuan Yang, Guile Wu, Yuan Ren, Kejian Lin, Bingbing Liu, Yang Liu, and Jinjun Shan. Vqa-diff: Exploiting vqa and diffusion for zero-shot image-to-3d vehicle asset generation in autonomous driving. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[36] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 7, 8

[37] Zhiyuan Ma, Yuxiang Wei, Yabin Zhang, Xiangyu Zhu, Zhen Lei, and Lei Zhang. Scaledreamer: Scalable text-to-3d synthesis with asynchronous score distillation. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[38] Gal Metzer, Elad Richardson, Or Patashnik, Raja Giryes, and Daniel Cohen-Or. Latent-nerf for shape-guided generation of 3d shapes and textures. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 3, 4

[39] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, 2020. 1

[40] Nasir Mohammad Khalid, Tianhao Xie, Eugene Belilovsky, and Tiberiu Popa. Clip-mesh: Generating textured meshes from text using pretrained image-text models. In *ACM SIGGRAPH Asia*, 2022. 2

[41] Alex Nichol, Heewoo Jun, Prafulla Dhariwal, Pamela Mishkin, and Mark Chen. Point-e: A system for generating 3d point clouds from complex prompts. *arXiv preprint arXiv:2212.08751*, 2022. 2

[42] OpenAI. Gpt-4 system card. *OpenAI*, 2023. 3

[43] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023. 7, 8

[44] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 4

[45] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021. 2

[46] ITU-R BT.500 Recommendation. Methodologies for the subjective assessment of the quality of television images. 2019. 4

[47] ITU-T P.910 Recommendation. Subjective video quality assessment methods for multimedia applications. 1999. 4

[48] Jiawei Ren, Liang Pan, Jiaxiang Tang, Chi Zhang, Ang Cao, Gang Zeng, and Ziwei Liu. Dreamgaussian4d: Generative 4d gaussian splatting. *arXiv preprint arXiv:2312.17142*, 2023. 1, 3

[49] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. In *Advances in Neural Information Processing Systems (NIPS)*, 2022. 2, 3, 7, 8

[50] Junyoung Seo, Susung Hong, Wooseok Jang, Inès Hyeonsu Kim, Min-Seop Kwak, Doyup Lee, and Seungryong Kim.

Retrieval-augmented score distillation for text-to-3d generation. In *International Conference on Machine Learning (ICML)*, 2024. 3

[51] Junyoung Seo, Wooseok Jang, Min-Seop Kwak, Hyeonsu Kim, Jaehoon Ko, Junho Kim, Jin-Hwa Kim, Jiyoung Lee, and Seungryong Kim. Let 2d diffusion model know 3d-consistency for robust text-to-3d generation. In *International Conference on Learning Representations (ICLR)*, 2024. 3

[52] Yichun Shi, Peng Wang, Jianglong Ye, Mai Long, Kejie Li, and Xiao Yang. Mvdream: Multi-view diffusion for 3d generation. *arXiv preprint arXiv:2308.16512*, 2024. 3

[53] Shaolin Su, Qingsen Yan, Yu Zhu, Cheng Zhang, Xin Ge, Jinqiu Sun, and Yanning Zhang. Blindly assess image quality in the wild guided by a self-adaptive hyper network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 7

[54] Jiaxiang Tang, Zhaoxi Chen, Xiaokang Chen, Tengfei Wang, Gang Zeng, and Ziwei Liu. Lgm: Large multi-view gaussian model for high-resolution 3d content creation. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[55] Jiaxiang Tang, Jiawei Ren, Hang Zhou, Ziwei Liu, and Gang Zeng. Dreamgaussian: Generative gaussian splatting for efficient 3d content creation. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 3

[56] Christina Tsalicoglou, Fabian Manhardt, Alessio Tonioni, Michael Niemeyer, and Federico Tombari. Textmesh: Generation of realistic 3d meshes from text prompts. In *International Conference on 3D Vision (3DV)*, 2024. 1, 2, 4

[57] Haochen Wang, Xiaodan Du, Jiahao Li, Raymond A. Yeh, and Gregory Shakhnarovich. Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 4

[58] Jianyi Wang, Kelvin CK Chan, and Chen Change Loy. Exploring clip for assessing the look and feel of images. In *AAAI Conference on Artificial Intelligence*, 2023. 7, 8

[59] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolificdreamer: high-fidelity and diverse text-to-3d generation with variational score distillation. In *Advances in Neural Information Processing Systems (NIPS)*, 2024. 2, 3

[60] Haoning Wu, Zicheng Zhang, Weixia Zhang, Chaofeng Chen, Liang Liao, Chunyi Li, Yixuan Gao, Annan Wang, Erli Zhang, Wenxiu Sun, et al. Q-align: Teaching lmms for visual scoring via discrete text-defined levels. *arXiv preprint arXiv:2312.17090*, 2023. 7, 8

[61] Tong Wu, Guandao Yang, Zhibing Li, Kai Zhang, Ziwei Liu, Leonidas Guibas, Dahua Lin, and Gordon Wetzstein. Gpt-4v(ision) is a human-aligned evaluator for text-to-3d generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 3

[62] Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023. 2, 3, 7, 8

[63] Zike Wu, Pan Zhou, Xuanyu Yi, Xiaoding Yuan, and Hanwang Zhang. Consistent3d: Towards consistent high-fidelity text-to-3d generation with deterministic sampling prior. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 4

[64] Jiale Xu, Xintao Wang, Weihao Cheng, Yan-Pei Cao, Ying Shan, Xiaohu Qie, and Shenghua Gao. Dream3d: Zero-shot text-to-3d synthesis using 3d shape prior and text-to-image diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[65] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. Imagereward: Learning and evaluating human preferences for text-to-image generation. In *Advances in Neural Information Processing Systems (NIPS)*, 2024. 2, 3, 7, 8

[66] Junkai Yan, Yipeng Gao, Qize Yang, Xihan Wei, Xuansong Xie, Ancong Wu, and Wei-Shi Zheng. Dreamview: Injecting view-specific text guidance into text-to-3d generation. In *European Conference on Computer Vision (ECCV)*, 2024. 3

[67] Runjie Yan, Yinbo Chen, and Xiaolong Wang. Consistent flow distillation for text-to-3d generation. In *International Conference on Learning Representations (ICLR)*, 2025.

[68] Xianghui Yang, Yan Zuo, Sameera Ramasinghe, Loris Bazzani, Gil Avraham, and Anton van den Hengel. Viewfusion: Towards multi-view consistency via interpolated denoising. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3

[69] Junliang Ye, Fangfu Liu, Qixiu Li, Zhengyi Wang, Yikai Wang, Xinzhou Wang, Yueqi Duan, and Jun Zhu. Dreamreward: Text-to-3d generation with human preference. In *European Conference on Computer Vision (ECCV)*, 2024. 3, 7

[70] Jianglong Ye, Peng Wang, Kejie Li, Yichun Shi, and Heng Wang. Consistent-1-to-3: Consistent image to 3d view synthesis via geometry-aware diffusion models. In *International Conference on 3D Vision (3DV)*, 2024. 3

[71] Taoran Yi, Jiemin Fang, Junjie Wang, Guanjun Wu, Lingxi Xie, Xiaopeng Zhang, Wenyu Liu, Qi Tian, and Xinggang Wang. Gaussiandreamer: Fast generation from text to 3d gaussians by bridging 2d and 3d diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 1, 3

[72] Taoran Yi, Jiemin Fang, Zanwei Zhou, Junjie Wang, Guanjun Wu, Lingxi Xie, Xiaopeng Zhang, Wenyu Liu, Xinggang Wang, and Qi Tian. Gaussiandreamerpro: Text to manipulable 3d gaussians with highly enhanced quality. *CoRR*, 2024. 3

[73] Biao Zhang, Jiapeng Tang, Matthias Nießner, and Peter Wonka. 3dshape2vecset: A 3d shape representation for neural fields and generative diffusion models. *ACM Transactions On Graphics (TOG)*, 2023. 2

[74] Longwen Zhang, Ziyu Wang, Qixuan Zhang, Qiwei Qiu, Anqi Pang, Haoran Jiang, Wei Yang, Lan Xu, and Jingyi Yu. Clay: A controllable large-scale generative model for creating high-quality 3d assets. *ACM Transactions on Graphics (TOG)*, 2024. 2

[75] Sixian Zhang, Bohan Wang, Junqiang Wu, Yan Li, Tingting Gao, Di Zhang, and Zhongyuan Wang. Learning multi-dimensional human preference for text-to-image generation.

In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3, 6

[76] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision (IJCV)*, 2022. 6

[77] Junzhe Zhu, Peiye Zhuang, and Sanmi Koyejo. Hifa: High-fidelity text-to-3d generation with advanced diffusion guidance. In *International Conference on Learning Representations (ICLR)*, 2024. 3